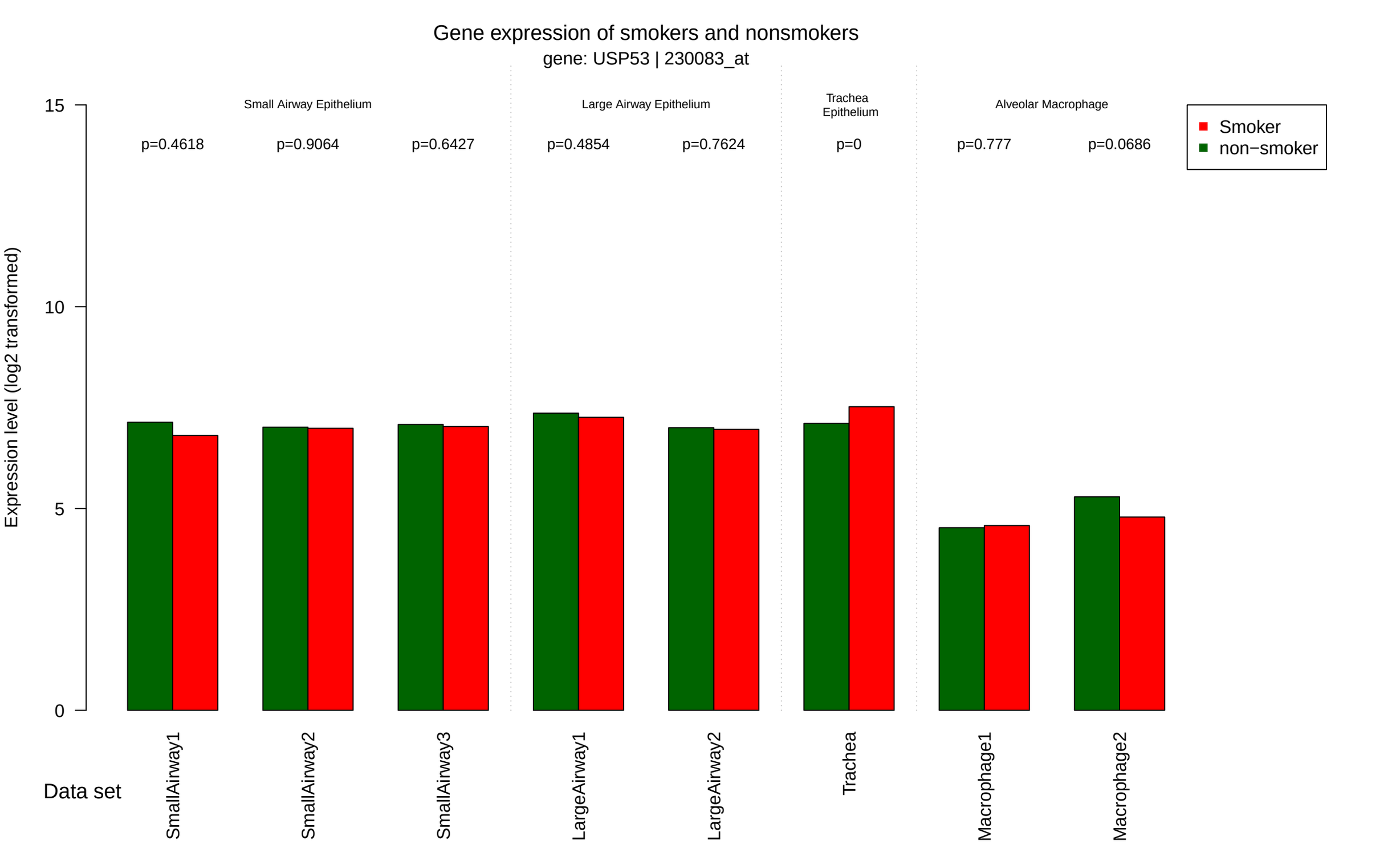
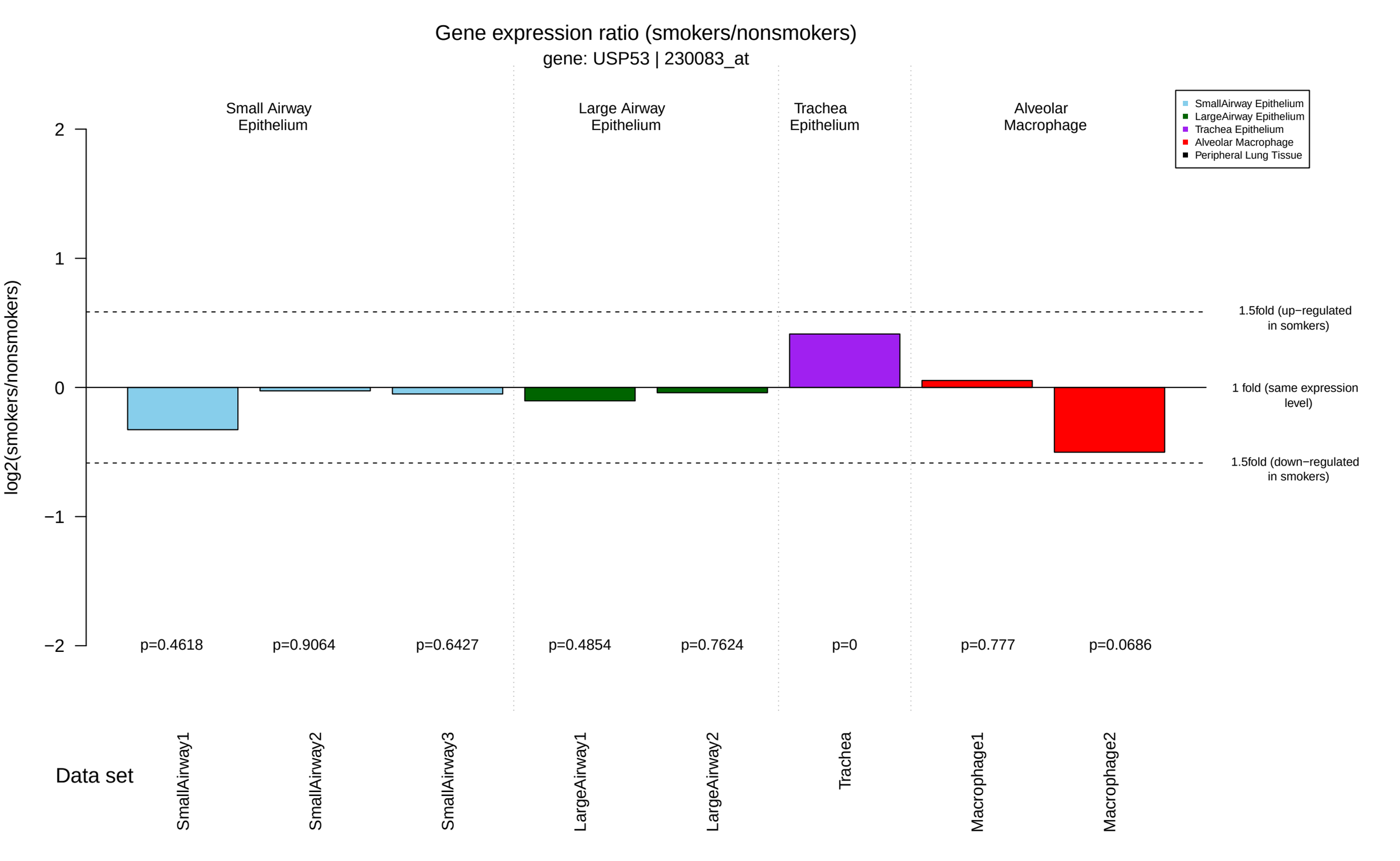
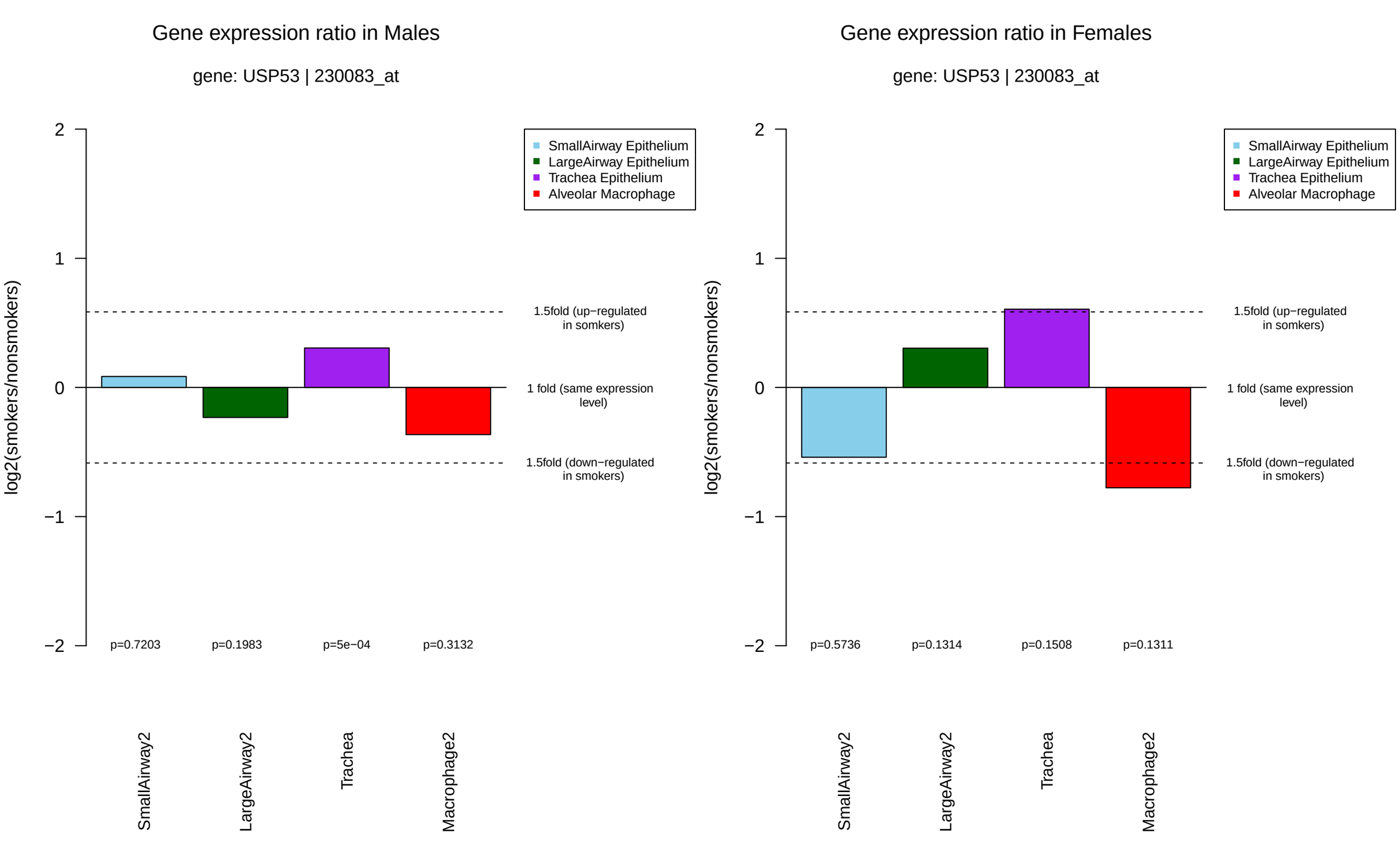
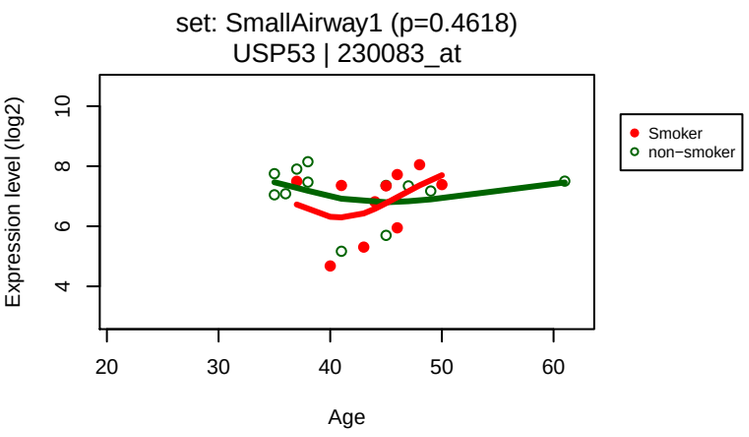
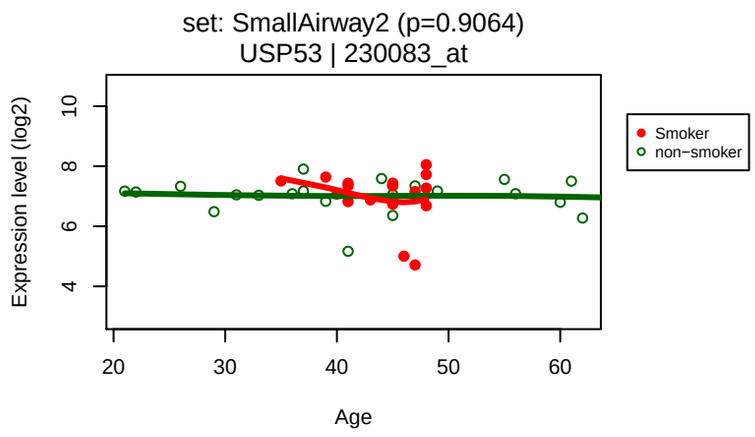
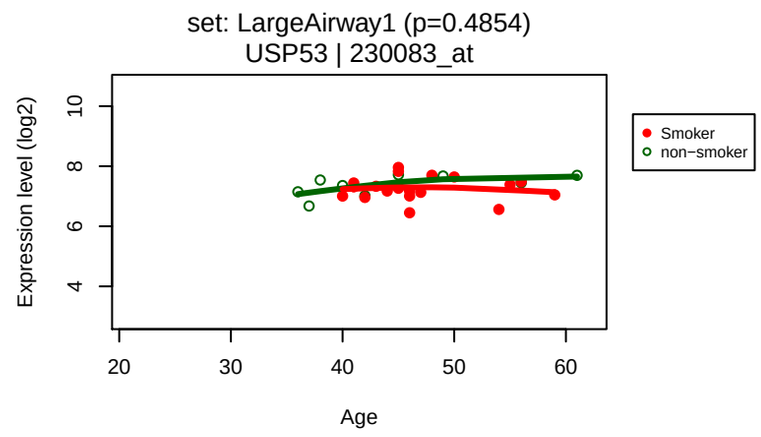

Gene expression of smokers and nonsmokers
gene: USP53 | 230083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4618
p=0.9064
p=0.6427
p=0.4854
p=0.7624
p=0
p=0.777
p=0.0686
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

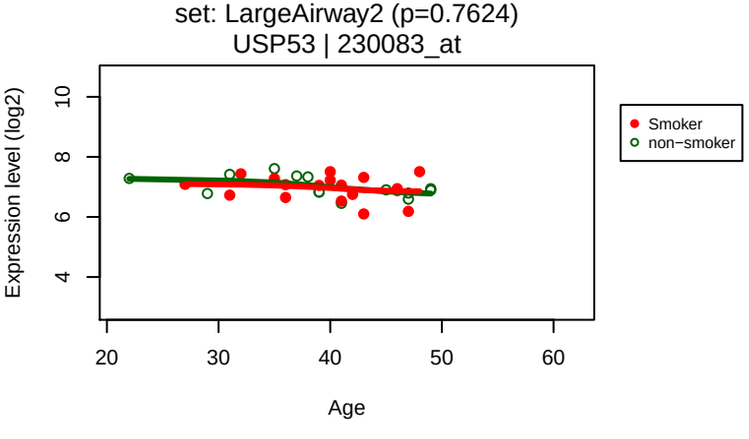

Gene expression ratio (smokers/nonsmokers)
gene: USP53 | 230083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4618
p=0.9064
p=0.6427
p=0.4854
p=0.7624
p=0
p=0.777
p=0.0686
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

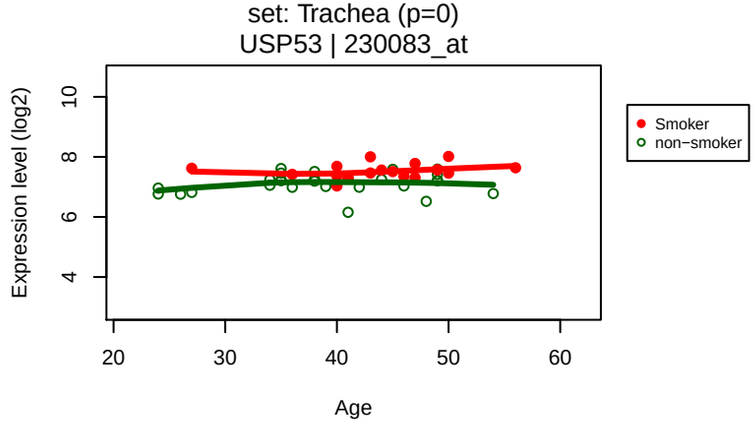

Gene expression ratio in Males
gene: USP53 | 230083_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7203
p=0.1983
p=5e−04
p=0.3132
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP53 | 230083_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5736
p=0.1314
p=0.1508
p=0.1311
SmallAirway2
LargeAirway2
Trachea
Macrophage2

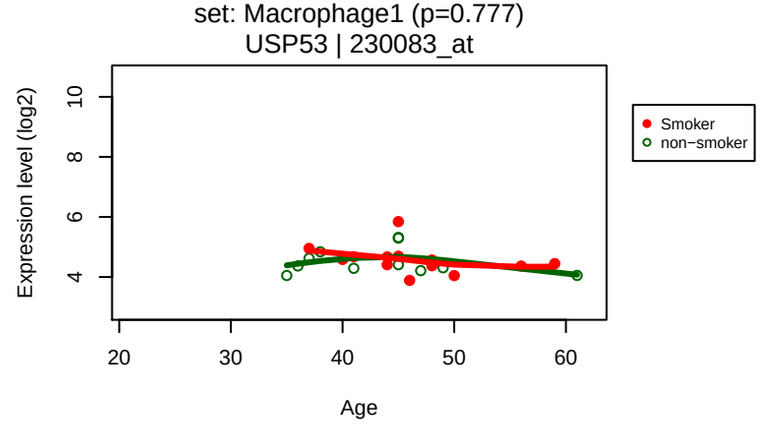

set: SmallAirway1 (p=0.4618)
USP53 | 230083_at
Expression level (log2)
Age
Smoker
non−smoker

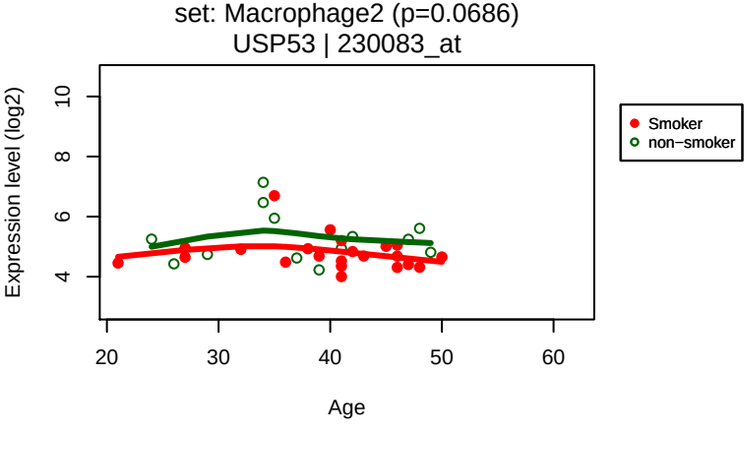

set: SmallAirway2 (p=0.9064)
USP53 | 230083_at
Expression level (log2)
Age
Smoker
non−smoker

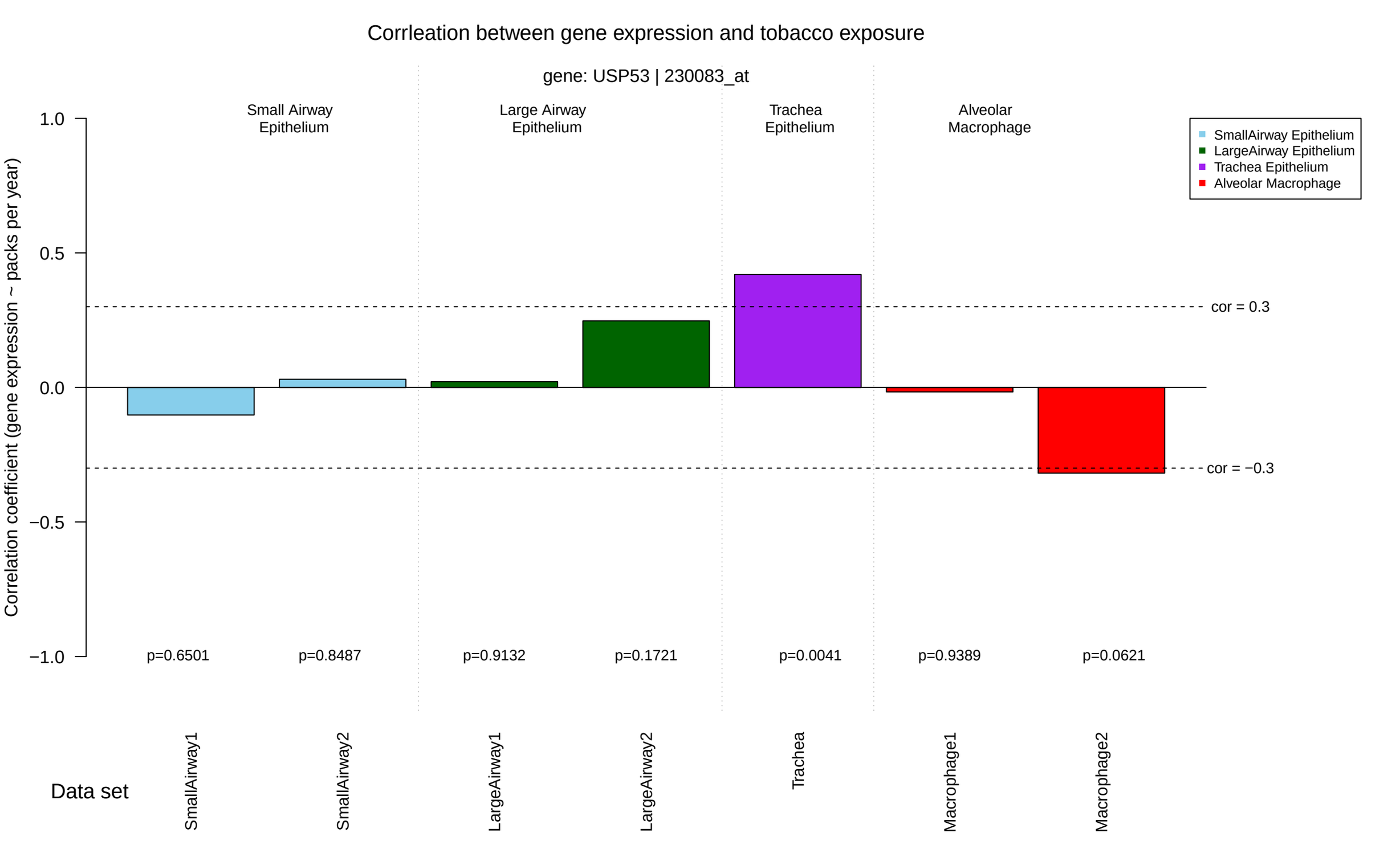

set: LargeAirway1 (p=0.4854)
USP53 | 230083_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7624)
USP53 | 230083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
USP53 | 230083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.777)
USP53 | 230083_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0686)
USP53 | 230083_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP53 | 230083_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6501
p=0.8487
p=0.9132
p=0.1721
p=0.0041
p=0.9389
p=0.0621
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2